

2021 Ph H2 Q14

Section: Electricity

Topic: Internal Resistance of a Cell

Experiment with cells to determine internal resistance.

(a) Describe how to use apparatus and analyse data to find r . (b)(i) Define EMF of 1.5 V. (b)(ii) 4 cells in series ($E=1.5$ V, $r=0.50$ Ω each), motor 20 Ω , ammeter reads 0.20 A: calculate R . (c) State and explain what happens to voltmeter reading when R is increased.

Worked solution

(a) Set up circuit with variable resistor, voltmeter across cell, and ammeter in series. Measure terminal potential difference V and current I for different values of load resistance. Plot V against I . The graph is a straight line with gradient $-r$ and intercept equal to EMF. Internal resistance r is determined from the gradient.

(b)(i) EMF of 1.5 V means the cell supplies 1.5 joules of energy per coulomb of charge when no current is drawn (open circuit).

(b)(ii) Four cells in series: total EMF = 6.0 V, total internal resistance = 2.0 Ω .

Total circuit resistance = $E_{\text{total}} / I = 6.0 / 0.20 = 30.0$ Ω .

Subtracting motor (20.0 Ω) and internal resistance (2.0 Ω): $R = 8.0$ Ω .

Answer: $8.0 \, \Omega$

(c) Increasing R decreases the current in the circuit. The lost volts (Ir across internal resistance) decreases, so terminal potential difference (the voltmeter reading) increases. Therefore, voltmeter reading rises towards the EMF of $6.0 \, \text{V}$.

Final answers

(a) Plot V - I graph, gradient = $-r$

(b)(i) EMF = energy per unit charge at open circuit

(b)(ii) $R = 8.0 \, \Omega$

(c) Voltmeter reading increases as R increases

Revision tips

- Use V - I graph to find internal resistance: slope = $-r$, intercept = EMF.
- EMF = work done per coulomb without current drawn.
- Series cells: total EMF = sum, total r = sum.
- Lost volts = Ir ; terminal PD = $E - Ir$.
- Increasing load resistance decreases current, reducing lost volts $\rightarrow$ terminal voltage rises.